

TITANIC SURVIVAL PREDICTION

CI345 Machine Learning Final Project

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PROBLEM STATEMENT

- **Binary Classification Problem**
- • **Predict if a passenger survived the Titanic disaster (yes/no)**
- • **Key features: sex, age, passenger Class**
- • **Historical dataset with social & economic insights**
- • **Goal: Build accurate classification models and gain meaningful insights**

DATASET OVERVIEW

- • **1,309 passengers (rows)**
- • **5 columns: names, survived (target), sex, age, passenger Class**
- • **Missing values: ~263 in age (~20%)**
- • **Target distribution: ~38% survived (imbalanced)**
- • **Data types: 1 numeric, 3 categorical**

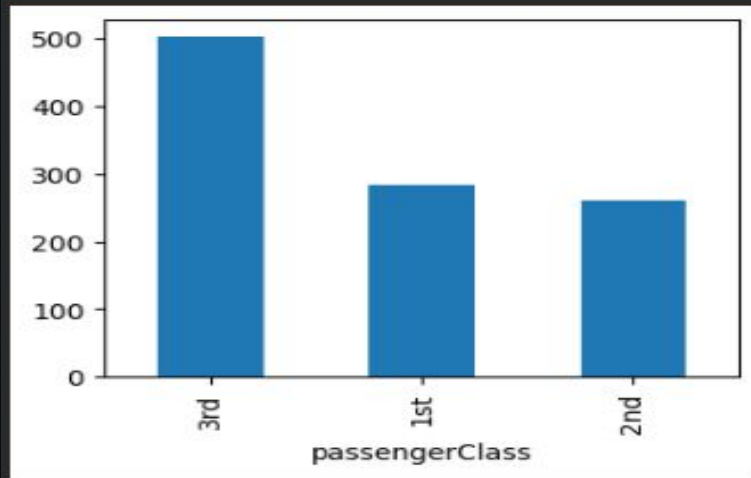
DATA FINDINGS

- **Key Insights:**
 - **Females: ~74% survival rate vs ~19% for males**
 - **1st class passengers had significantly higher survival**
 - **Children & women are prioritized ("women and children first")**
 - **Sex and passenger Class are the strongest predictors**

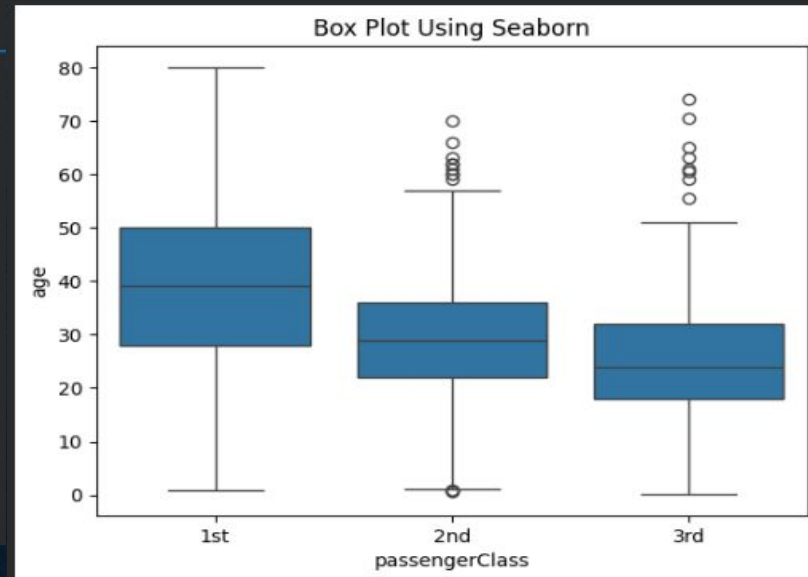
VISUALIZATIONS

- Important Visualizations:
- Survival Rate by Passenger Class (Pie Chart)
- Survival Rate by Passenger Class (Bar Chart)
- Age Frequency (Histogram)
- Age Distribution by Passenger Class (Box plot)

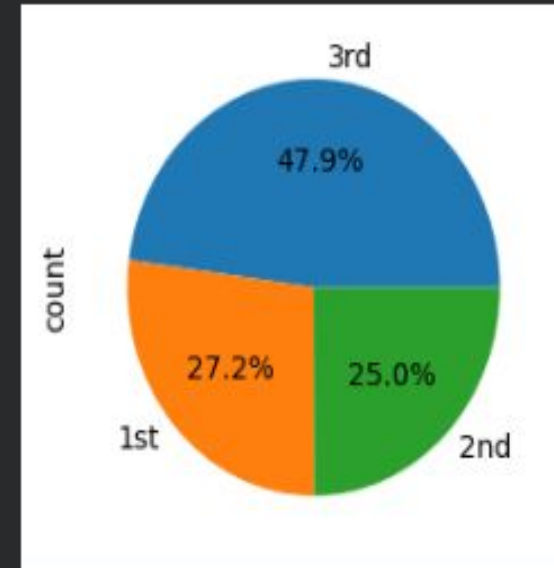
```
res = df_copy.passengerClass.value_counts()  
res.plot(kind='bar', figsize=(4,3))  
plt.show()
```



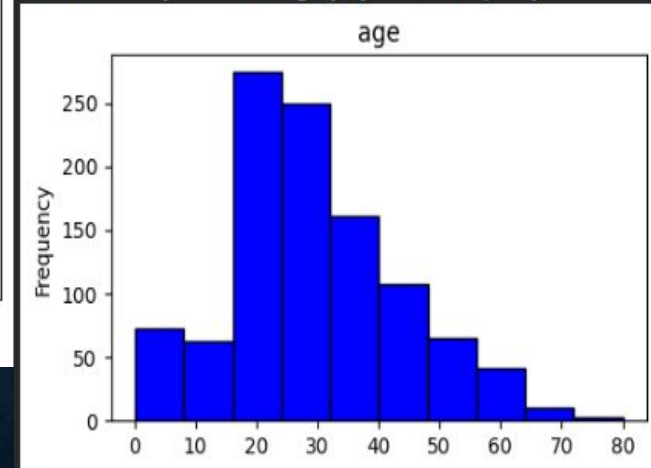
```
import seaborn as sns  
  
sns.boxplot(data=df_copy, x="passengerClass", y="age")  
  
# 3. Display the plot  
plt.title("Box Plot Using Seaborn")  
plt.show()
```



```
res = df_copy.passengerClass.value_counts()  
res.plot(kind='pie', figsize=(4,3), autopct='%1.1f%%')  
plt.show()
```



```
res = df_copy['age']  
res.plot(kind='hist', title = 'age', edgecolor='black',color= 'blue', figsize =( 5,3))  
  
<Axes: title={'center': 'age'}, ylabel='Frequency'>
```



DATA PREPROCESSING

- • **Dropped irrelevant "names" column**
- • **Label Encoding: sex (male=0, female=1)**
- • **Label Encoding: passengerClass**
- • **Median imputation for missing age values**
- • **Train-test split (70/30)**

CLASSIFIERS

- **3 Classification Algorithms:**

- 1. K-Nearest Neighbors (KNN) - $k = 3, 5, 7$
 - Tested with $k = 3$, $k = 5$, and $k = 7$
 - Best result with $k = 5$ (Accuracy $\approx 74\%$)
- 2. Decision Tree Classifier
 - Used default parameters with random state being 42
- 3. Linear Discriminant Analysis (LDA)

RESULTS

- **Model Performance:**
 - **KNN (k=5): Accuracy $\approx 74\%$**
 - **Decision Tree: Accuracy $\approx 73\%$**
 - **LDA: Accuracy $\approx 72\%$**
- **More information about the classification reports & confusion matrices are available in the code**

COMPARISON

Model Comparison Table:

• Model	Accuracy	Precision	Recall	F1
• -----	-----	-----	-----	-----
• KNN	0.74	~0.74	~0.73	~0.73
• Decision Tree	0.73	~0.73	~0.72	~0.72
• LDA	0.72	~0.71	~0.71	~0.71
• KNN had best results overall				

CONCLUSIONS

- • Sex and passenger class are shown to be key survival factors
- • The classification models successfully captured historical patterns
- • This aligns with known Titanic history ("women and children first")
- • Future work: Ensemble methods, additional features
- • Thank You! Questions?